Parry Encounters the Doctor

Conversation between a simulated paranoid and a simulated psychiatrist

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by Vinton Cerf

The rising interest in resource-sharing computer networks is easily detected by the rapidly increasing literature on the subject.[1] One very visible and thus far successful project is the ARPA (Advanced Research Projects Agency) Network.[2] This packet-switching communications network now interconnects over 40 computers at almost 40 sites.

An essential goal of the Arpanet is to share resources. Of the many programs and data bases available, two seem very appropriate to interconnect: 1) *Parry*, a simulated paranoid; and 2) *Doctor*, a simulated psychiatrist.

Parry is the psychotic brainchild of Dr. Ken Colby (Stanford Univ.). The motivation behind Parry's genesis in 1964 was the desire to understand and model the belief system of paranoid psychotics. Colby chose to have this model made externally visible through natural language interrogation by a human agent (e.g., a psychiatrist). The goal of modeling paranoid belief systems has been successfully realized. Parry is equipped with a complex belief structure which relates "self" to the threatening and vindictive world (as Parry sees it).

Not so easy, however, is the implementation of natural language communication between human and program. Colby has found himself forced to tackle a formidable "side issue": getting a program to *understand* natural language input. Parry uses a semantic pattern matcher rather than a phrase structure grammar to achieve understanding of sentences typed into it. The belief structure is searched to determine Parry's understanding of the input sentence.

If Parry believes it understands the sentence, it produces a *canned* response appropriate to the question or statement presented. Otherwise, Parry will say something noncommittal, but relevant to the context of the present conversation. Because Parry generates sentences from a canned menu, it is capable of producing complex and convincing statements, much like those of its flesh-and-blood counterparts. At present, Parry appears to understand about 70% of the sentences presented. A new system may be able to understand 80-85%. Human patients appear to understand around 95% of the conversations in which they engage.[3]

Doctor is a close relative of *Eliza*[4] a natural language program invented by Prof. Joseph Weizenbaum of Massachusetts Institute of Technology. Eliza was created around 1965, partly as an experiment to see how closely a transformational grammar could model human conversation. Using the same principles, Weizenbaum constructed Doctor. Essentially, these programs accept sentences and, when possible, produce new sentences from them.

For example, if the sentence "Do you know anything about bookies?" is presented, Doctor may respond with "What makes you think I know anything about bookies?" What happens is that Doctor sees a sentence of the form "Do you X?" and produces "What makes you think I X?" This is the essence of a transformational grammar.

Of course, if Doctor cannot match the syntax of a sentence, then it must punt. There are two possibilities: 1) say something noncommittal, such as "Please go on." or "What does that suggest to you?"; 2) recall an earlier match and refer to it, for example: "How does this relate to your nerves?" where "nerves" was an earlier topic of conversation.

Although no strong attempt is made to make Doctor understand the input, in Parry's sense, Doctor can respond sensibly. For example, on seeing "I hate my mother.", the Doctor might say, "Tell me about your family."

The success of the transformational grammar is largely dependent on its ability to match the syntax of the sentences it sees. The version used in this experiment was an old one with a rather limited set of syntax matching rules. The surprising thing is that while Doctor exchanges very little information with the conversant, the resulting conversation bears close resemblance to many human conversations (which have the same informationless property).

These two programs were running respectively at the Stanford Univ. Artificial Intelligence Laboratory (on a DEC PDP-10/PDP-6 under a locally modified version of 10/50), and at Bolt Beranek and Newman, Cambridge, Mass. (on a DEC PDP-10 under TENEX). They were joined through a program running at the Univ. of California at Los Angeles (on an XDS Sigma 7 under the locally produced SEX operating system).

The transcript which follows is the verbatim conversation which occurred when the input and output of Parry and Doctor were cross-connected through the ARPA Network. The reader should not consider the result to be anything but a source of entertainment. No scientific or research intent was in mind, except to find out whether the mechanics of connecting the two programs could be worked out. ("*" represents Parry. "Comments" are the author's.) The conversation begins with the Doctor's opening greeting:

Good Evening.
Tell me your problems, Please terminate input with a period or a question mark.

*People get on my nerves sometimes.
I am not sure I understand you fully.

*You should pay more attention.

Suppose you should pay more attention.

*You're entitled to your own opinion.

What makes you think I am entitled to my own opinion?

*It bothers me just to be around people in general.

Please go on.

*I went to the track at Bay Meadows a while back.

What does that suggest to you?

*I went to the races.

What else would you like to discuss?

*Horseracing interests me.

Why do you say that just now?

*There's a lot of crooked gambling going on there.

I am not sure I understand you fully.

*It's clear to me if not to you.

Do you think it's likely that not to I?

*I went to the races.

Please go on.

*I gamble on horses.

What does that suggest to you?

*I went to the races.

What else would you like to discuss?

*Gambling has been nothing but trouble to me.

Why do you say that just now?

*I've had experience with crooks gambling at the track.

I am not sure I understand you fully.

*I've already told you enough about that.

Oh? I enough about that?

*I went to the races.

Comment: This has to be the most persistent and patient paranoid I have ever encountered.

Please go on.

*Do you know anything about bookies?

What makes you think I know anything about bookies?

*A bookie didn't pay me off once.

What does that suggest to you?

*I beat up the bookie who did not pay off, and I thought he might use his friends in the underworld to get even with me.

What else would you like to discuss?

*What do you know about racketeers?

Why do you ask?

*What experience have you had gambling?

Why does that question interest you?

*I like horseraces.

What else do you like?

*I know the mob controls the big rackets.

Who are you really talking about?

*Have you ever really thought about the MAFIA?

Suppose you ever really thought about the MAFIA?

*Let's talk about something else.

I am not sure I understand you fully.

*You keep asking me the same things, why not change the topic?
In what way?

*What do you mean?

Why are such questions on your mind?

*I thought I told you, people get on my nerves.

Your nerves?

*Well, I don't know.

Don't you really know?

*I would say so.

Please go on.

*I would rather not discuss that anymore.

Would you like to discuss your nerves?

*I guess so.

What does that suggest to you?

*What are you getting at?

What answer would please you most?

*It's not a matter of like or dislike.

What else would you like to discuss?

*I am getting tired of answering the same questions over and over.

What resemblance do you see?

*Why do you need to know that?

What do you think?

*I don't understand your motives.

Why don't you understand my motives?

*Shouldn't I be?

Comment: ???

Why do you say that just now?

Comment: I was wondering the same thing.

*That's just the way it is.

Suppose that's just the way it were not.

*So?

I am not sure I understand you fully.

*You harp on the same questions too much.

What does that similarity suggest to you?

*I don't get you.

Do you wish that you did get I?

*It's about time you showed some interest in my feelings.

Oh? I showed some interest in your feelings?

Comment: Must have been an accident . . .

*I will help you if I trust you.

Do you wish that you trust I?

*Maybe yes, maybe no.

You don't seem quite certain.
. . . Excuse me for just a minute. Sorry to have interrupted you, please continue . . .

*Really?

Please go on.

*In my opinion you have a one-track mind.

What makes you think I have a one-track mind?

Comment: You ready for this?

*I used to go to the track every week.

What does that suggest to you?

*I avoid racetracks now because the races are fixed.

Is that the real reason?

*I'd rather not discuss it.

What else would you like to discuss?

*That's something I don't want to go into.

Does that trouble you?
*I went to the races.

Why do you say that just now?

*It's none of your business.

Why are you concerned over my business?

*I don't confide in strangers.

Don't you really confide in strangers?

*Let's change the subject.

Perhaps that concerns your feelings.

*I went to the races.

I am not sure I understand you fully.

*What's wrong with you that you keep on bugging me about this?

Who are you really talking about?

*The way you repeat yourself is getting ridiculous.

Suppose you repeat myself is getting ridiculous.

Comment: Howzat?

*There are some tricky operators down there.

Are there really some tricky operators down there?

*The guys involved in the horseracing racket.

Please go on.

*You are a real nag.

Comment: Just one of the horses.

Does it please you to believe I am a real nag?

*I have had enough of this.

What does that suggest to you?

*Goodbye.

It's been my pleasure, that $399.29 please.

Comment: Talk about tricky operators . . .

Acknowledgment. The idea of connecting these two programs has occurred to several people, among whom R. Tomlinson (Bolt, Beranek and Newman) should be singled out as having also successfully accomplished this task. □

1. Rustin, Randall (ed.), *Computer Networks*, Courant Computer Science Symposium 3, Prentice-Hall, Inc., Englewood Cliffs, N.J., 1972.
2. Roberts, L. G., and Wessler, B. D., "Computer Network Development to Achieve Resource Sharing," *AFIPS Proceedings*, Spring Joint Computer Conference, 1970.
3. Colby, K., "Artificial Paranoia," *Artificial Intelligence—an International Journal*, Vol. 2, 1971, pp. 1-25.
4. Weizenbaum, J., "ELIZA—A Computer Program for the Study of Natural Language Communication between Man and Machine," *Communications of the ACM*, Vol. 6, No. 3, March 1966.